{Lec1}

1-Many courses in computer science and electronic engineering focus on the technical

2-Technical Skills: These include programming, modeling, testing, and requirement specification.

3-Process Skills: These define when, how, and in what order those technical tasks should occur.

4-Writing good requirements is no good if you don't have a controlled process.

5-The techniques to do testing are useless if you don't have any time left to do testing.

6- Software project management is interesting and challenging because the product is intangible.

7- The software product is uniquely flexible, allowing different sizes and constraints.

8- Many software projects are considered one-off, meaning they are custom-built solutions.

9- A one-off project is designed to meet the specific needs of client a or organization

10- The size and complexity of software systems are increasing exponentially.

11- Human lives might depend on software running as expected (consider the control system of an airplane)- safety critical systems

12- Initiation and Approval– before the SDLC even starts

13- Closing – after the SDLC completes

14- Maintenance → occurs after deployment, not formally part of the PMLC.

(note)15- The project management life cycle encompasses all the activities of the project, while the systems development life cycle focuses on realizing the product requirements.

16- Project Initiation: Starts when an organization recognizes a need for a new or improved capability and explores how to address it.

17- Project Charter: A concise document that outlines project objectives, scope, and responsibilities to gain approval from key stakeholders.

18- KPIs :Measurable values that demonstrate how effectively a project or business is achieving its objectives.

KPIs→Key Performance Indicators

19- Forecasts: Predictions about future project outcomes based on current data, trends, and assumptions.

20- Post Mortem: A review conducted after a project is completed to assess its successes, challenges, and lessons learned.

21- Project Punchlist: A list of remaining tasks or issues that need to be addressed before a project is considered complete.

22- Reporting: The process of regularly communicating project progress, status, and key metrics to stakeholders.

Project initiation→23-Deliverable: Project Initiation Document (PID) that outlines the project's goals, business need, stakeholders, and high-level requirements

Project planning→24- project plan that includes timeline, resources, budget, scope, and risk management strategies.

Project planning→25- Scope Statement: Clearly defines the website's features.

Project planning→26- Work Breakdown Structure (WBS): Breaks the project down into smaller tasks.

Project planning→27- Risk Management Plan: Identifies risks like delays in content creation or issues with payment integration

Project planning→28- SMART Goals:"Complete website design by the end of month 2."

29- Project Execution: Implement tasks such as developing the user interface, coding, and integrating the backend.

Project execution→30- Kick-off meeting: Align the team on project expectations.

Project execution→31- Team Development: Assign roles like web developers, designers, and content writers.

Project execution→32- Updates and Progress Reports: Regular status reports that detail work completed.

33- Project Initiation: Create a project charter and business case.

34- Project Performance and Monitoring: Use KPIs to track progress, such as checking if development is on schedule.

35- Performance Tracking: Regular reports to monitor progress against the timeline and budget.

36- Issue Resolution: Address any delays or problems that arise, such as a delay in coding due to unanticipated technical issues.

37- Project Close: Complete all tasks, perform a final project review, and deliver the website to the client.

38- Final Budget and Report: Provide a final account of all expenses and report on whether goals were met.

39- Project Punchlist: Resolve any remaining tasks (e.g., fixing minor bugs).

40- Final Deliverable: Fully functional website launched to the public.

(note)41- First used to schedule and monitor work and progress in ship building

42- CPM: Mathematically based algorithm for scheduling a set of project activities, used to plan maintenance activities in plants.

CPM→ Critical Path Method

{Lec2}

43-A project is a temporary endeavor undertaken to create a unique product, service ,or result.

44-Temporary Definitive begin and end.

Note→45-Projects’ results are not necessarily temporary (used long-term by the client)

46-Resource constrained A limited time/ resources is available to build the project outputs.

47-A product which is quantifiable (e.g. a software app).

48-A capability– A result to perform a service, such a business function (payroll system).

49-A result such as knowledge (collected in documents, presentation).

50-Progressive elaboration: Development by steps and in increments

Note→51-The cost of changes increases as a project progresses

52-Unique output: A project delivers has some novelty.

53-Product Development [tangible]: is a type of software systems i.e., outcome of software developing process and are delivered to customer along with documentation.

54-Service [intangible]: Involves providing software as a service (SaaS) over the internet.

55-An initiating phase, during which the project infrastructure and the project's goals are drafted.

56-in initiating phase, defining what the project is and gaining approval to move forward.

57-A planning phase, during which project goals are refined, activities identified and scheduled, and many other support activities are properly planned.

58-In planning phase, defining how the work will be done by setting clear goals, timelines, budgets, and resources.

59-An executing phase, during which the actual work takes place.

60-a monitoring phase measures the progress and raises flags when plans and reality disagree.

61-A final closing phase, where the project outputs are handed out and the project is closed.

62- A program is a set of related projects managed in a coordinated way.

63-Subprojects: Projects may be divided in subprojects.

64-A portfolio refers to a grouping of unrelated projects, and programs.

65-The purpose of a portfolio is to establish centralized management and oversight for many projects and programs.

66-Project and Program Management: Set of related projects managed in a coordinated way in order to achieve some sort of benefit.

67-Portfolios and Portfolio Management : Collection of unrelated projects or programs and other work grouped together to facilitate management and meet strategic objectives.

68-Projects and operations are both parts of the product life cycle.

69-Running restaurant service is a example of operations

70-One-offs: (bespoke systems) systems specifically created for a client.

71-Off-the-shelf: to fill the need of a large set of users.

72-Customized off-the-shelf: standardized systems which require a significant amount of customization to be used in an organization.

73-ERP is a example Customized off-the-shelf

ERP→ Enterprise Resource Planning

74- ERP: is made up of integrated modules or business applications that talk to each other and share common a database.

Note→75- Each ERP module typically focuses on one business area

76- Process and Systems Re-Engineering: change the way in which the operational work of an organization is carried out to achieve some strategic goal.

77- System Integration Services: automating the information flow among the systems of an organization.

78- System Integration Services: The approach is chosen when migrating to a new system is impractical or too costly.

79- Vertical : integration of systems performing similar operations.

80- Horizontal : integration of systems automating different steps of a procedure.

81- Consulting Services: Typically asked to gain a know-how outsize a company’s core competence.

82- Consulting services are expert services provided to organizations when they need specialized knowledge or skills that are not part of their core business.

83- The open-core model is a business model for the monetization of commercially produced open-source software.

84- A project stakeholder is any individual or an organization that is actively involved in a project.

85- The project manager: has the responsibility of project planning, procuring, and executing the project.

86- The project team: responsible for carrying out the work in a project

87- The project sponsor: responsible for approving, funding, and ensuring the success of a project, even without direct management authority.

88- The performing organization: is the primary group that is most directly involved in the project.

89- The client: benefits from the project outputs.

90- The“rest”: anyone who might be affected by the project outputs.

91-an internal stakeholder is anyone who has a direct interest in you or your organization.

92-an external stakeholder is a person or organization who has an interest in the success or failure of a project.

93- Software project management is the integration of management techniques to software development.

{Lec3}

94-user stories using this pattern: As a [user] I want to do [this] because of [that].

95-Diagrammatic a set of diagrams (e.g., descriptions use case diagrams) and corresponding textual

96-List of Requirements→a set of requirements written in natural language (plain English) describing system functions and properties in free text form.

97-User Stories: Structured textual descriptions of user functions

98-Use Case Diagrams→Diagrams describing the interaction between users and the system

99- Use Case Diagrams →Textual description of the interaction as a sequence of steps

100- A UML use case diagram summarizes the interactions between a system and its users.

101- Extend relationship: The use case is optional and comes after the base use case.

102- Include relationship: The use case is mandatory use case and part of the base use case.

Note ↓

Parameters	User Story	Use Case
Descriptions	The user story contains simplified and short descriptions.	A use case contains complete and lengthy descriptions.
Purpose	It is used to capture the requirements of a project in a simpler and more concise way.	It is designed to provide a detailed description of how a system should work.
Ease of Use	Users may read user stories with ease.	Describes in detail the interaction between users and the system.
Type of Interaction	It provides a single description.	It contains the sequence of interactions.
Management	It is manageable in one sprint.	It spans many sprints.
Time requirement	It is typically written in one or two sentences and is easy to read and understand.	It is usually much longer than User Stories, as they provide a more detailed description of how a system should work.

Note→Waterfall Model (linear approach) / Agile Methodology (iterative development)

103- Explicit assumptions can be validated during analysis.

104- implicit validation is carried to design and implementation.

105-Inconsistencies→ when different parts of the requirements behave in contradictory ways.

106-Incompleteness→arises when the behavior of the system is not specified for certain situations

107-Duplicates→when the same requirement is described multiple times.

108-(ERP)→ are systems that can automate integrating and simplify the processes of an organization, the data and the procedures of different business units.

ERP→Enterprise Resource Planning

109-An important part of the project work will focus on Mapping the current procedures to fit the new system.

110-The current procedures are often mapped to processes supported by which type of system? A→ERP

111-Business process modeling models the way in which an organization works.

112-Business process re-engineering plans the way in which an organization works, to make its operations more efficient.

113-''as is'' analysis describes the organization before the introduction of a new system

114-"to be" phase defines how the organization will change with the introduction of the new system

115-Business Process Modeling →Articulated and complex , it is sometimes planned and organized as an independent project.

116-Shadow processes are business tasks and workflows that are typically not formally tracked or monitored by the organization.

117-Organizational structure: chain of responsibility and accountability identify the changes that will have to be implemented in the organization to support the new processes

Note→responsibility [task-focused]

Note→accountability [results-focused]

118-Business processes: how an organization carries out its procedures – flow diagrams sketched.

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Aspect	Component Diagram	Deployment Diagram
Purpose	Represent the high-level software structure	Model the physical deployment of software components
Focus	Logical organization and relationships of components	Physical deployment on hardware and nodes
Main Elements	– Components (e.g., classes, packages, subsystems)	– Nodes (e.g., servers, workstations)
	– Interfaces (e.g., contracts, APIs)	– Artifacts (e.g., files, databases)
	– Connectors (e.g., associations, dependencies)	– Associations (indicating deployment relationships)
Use Cases	– Design phase: System architecture and structure	– Implementation phase: Deployment planning
	– Illustrate component relationships and interfaces	– Visualize component distribution on hardware
Abstraction Level	High-level abstraction	Low-level abstraction
Notation	Components, interfaces, connectors	Nodes, artifacts, associations
Relationships Represented	Dependencies, associations, aggregations, etc.	Deployment associations, mapping of components to nodes
Example Scenario	Representing software modules and their interactions in a banking application	Visualizing how web server software components are deployed on physical servers

119-Requirements Structuring Goal Improving maintenance of requirements over time

120-UserExperience Design goal Providing a coherent and satisfying experience on the different artifacts that constitute a software system, including its design, interface, interaction, and manuals

121-User-centered analysis: understanding how users will interact with the system (focus groups, experiments, surveys)

122-User-centered design: specifying how users will actually interact with the system (mock-ups)

123- wireframe is a quick sketch of a product intended to convey its desired functionalities

124- mockup is a realistic design of a product designed to gather feedback on its visual elements

125- prototype is an interactive simulation of a product designed to test the user experience.

{Lec4}

126-System Design Goal Defining the structure of the software to build (= system architecture)

127-Monolithic Architecture: A single codebase containing all components and functions.

128-Microservices Architecture: Separate services responsible for individual functions, each with its own lifecycle.

129- note→-A Pipeline Architecture -Layered Architecture

-Data-Centric -A client-server Architecture

130-Layered/Hierarchical organized into multiple control levels.

131-At the lower level , sensors are responsible for reading data from the environment

132-At the higher level , a controller takes the input of the sensors and decides the action to perform

133-model→how data are to be stored

134-view→how data are to be presented to the user.

135-controller→logic of the operations

136-Client-server→Focus on communication protocolo/service specifications

137-• Verification = did we build the system right?

138-Validation = are we building the right system?

139- Unit testing– a piece of code, such as aclass – automated testing

140-Integration testing→the interaction between two components

141-System testing→the system behaves as expected and implements correctly all the requirements

142-Usability testing→running in parallel with the other testing activities

143-human factor: is the people ready to use the system

144-data factor: is all the data which is needed for the system to run available to the new software

145-hardware factor: are all interfaces ready and functional

146-Cut-over: the new system completely replaces the old one.

147-Piloting: the new system is installed for a limited number of users

148-PhasedApproach: functions are rolled out incrementally.

149-Operations and Maintenance→the goal is Ensuring the system runs smoothly after its release.

[Types of Maintenance]

150-Corrective→if relative to fixing an issue discovered after the release of the system

151-Preventive→if relative to fixing an issue discovered, but not occurred

152-Adaptive→if relative to adapt a system to changed external conditions.

153-Perfective→if relative to improve some characteristics of a system.

154-Deployment Goal→Installing the new system and making it operational.

155-Code churn→measures how frequently code changes occur

156-a bug tracking system is used to manage and track tickets

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{Lec5}

157-Internal Development (Make) – Developing the project in-house using your organization’s resources, staff, and infrastructure.

158-External Development (Buy) – Getting the project from outside the organization instead of building it internally.

159-SaaS (Software as a Service) → subscribe and use online.

160-Off-the-shelf software → buy and install locally (one-time license)

161-Outsourced/custom-built → external company builds it for you

162- Strategic alignment means project matches organizational goals

163- Internal software is built for employees to improve internal processes (e.g., HR or payroll systems).

164- External software is developed for customers or the market to generate revenue

165- Sustainability in software engineering refers to The capacity of sustaining the project and its outputs after the project end

166- Required resources→ What the project needs to be completed

167- Current workload and availability→ How busy the team is now and whether they have time to take on new work.

168- Future demand and capacity→Expected future projects and whether the organization

169- Timing→deliver too early or too late and the outputs of the project might be useless

170- Technical Difficulty and Uncertainty→ the actual capability of solving various technical challenges, when the time comes

171-Profit = Returns – Investments

172-Annual Profit = Profit / Duration

173-ROI = Annual Profit / Investments

174- $$\text{Discount Factor} = \frac{1}{(1+r)^t}$$


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